

Entity Types & Entity Sets

Q. Define Entity, Entity Type, and Entity Set. Clearly differentiate between a Strong Entity and a Weak Entity with examples.

[DEF] Definition to Remember

An Entity is a single real-world object that has independent existence and can be uniquely identified. An Entity Type is a category of entities sharing the same attributes (the blueprint), and an Entity Set is the actual collection of all instances of that entity type stored in the database.

1. Entity, Entity Type, and Entity Set

- * Entity: A single, specific, real-world object that can be uniquely identified.
- * Entity Type: A category/template for entities sharing the same attributes. Represented by a Rectangle in ER diagrams.
- * Entity Set: The actual collection of all entities of a particular type in the database at a given moment.

2. Strong Entity vs. Weak Entity

Entities are classified based on their ability to be uniquely identified using their own attributes.

Strong Entity

Weak Entity

[EMPLOYEE]	<< DEPENDENTS >>	[[DEPENDENT]]
(Strong)	Double Diamond	(Weak)
Single Rect		Double Rect

3. Summary Table

Feature	Strong Entity	Weak Entity
Primary Key	Has its own Primary Key	Has only a Partial Key (Discriminator)
Existence	Independent	Depends on a Strong Entity
ER Symbol	Single Rectangle	Double Rectangle
Relationship	Single Diamond	Double Diamond (Identifying Relationship)
Example	`EMPLOYEE` (identified by `Emp_ID`)	`DEPENDENT` (identified by `Emp_ID + N`)

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Entity = single specific object; Entity Type = category template; Entity Set = all instances in DB.
2. Entity Types are represented by Rectangles in ER diagrams.
3. Strong Entity has its own Primary Key and exists independently.
4. Weak Entity has no Primary Key -- uses a Partial Key + Owner's PK for identification.

5. Weak Entity is represented by a Double Rectangle; its relationship by a Double Diamond.

6. If the Strong (Owner) Entity is deleted, the associated Weak Entity must also be deleted.

7. Example: EMPLOYEE (strong), DEPENDENT (weak -- relies on Employee_ID).

[RECALL] Quick Recall

Entity (object) -> Entity Type (blueprint/Rectangle) -> Entity Set (all instances) -> Strong (own PK, single rect) -> Weak (partial key, double rect, double diamond)